

# Emir Ceyani

Ph.D. Candidate in Electrical & Computer Engineering, University of Southern California

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## Research Interests

- Large Language Models
- Graph Neural Networks
- Generative Adversarial Networks
- Flow-Based Generative Models
- Conformal Prediction
- Uncertainty Quantification
- Federated Learning
- Bayesian Methods
- AI For Science & Engineering

## Education

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| <b>University of Southern California</b> , PhD in Electrical & Computer Engineering<br><i>Thesis: Recent Advances in GNN Training: Federation, Personalization, and Applications.</i> | <b>Jan 2021 – Jun 2026</b><br><i>Los Angeles, CA</i> |
| <b>Bilkent University</b> , MSc in Electrical Engineering<br><i>Thesis: Spatiotemporal Forecasting over Graphs with Deep Learning</i>                                                 | <b>Jun 2018 – Dec 2020</b><br><i>Ankara, Türkiye</i> |
| <b>Bilkent University</b> , BSc in Electrical Engineering                                                                                                                             | <b>Sep 2013 – Jun 2018</b>                           |

## Research Experience

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| <b>University of Southern California</b> , Graduate Research Assistant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Jan 2021 – Present</b>  |
| <ul style="list-style-type: none"><li>• <b>Federated Graph Learning:</b> Proposed the problem of training of graph neural networks in a federated setting and released the first benchmarking study to the literature. Improved personalization of federated graph neural networks (AAAI'22, SIAM-SDM'25.)</li><li>• <b>AI-Based Analog &amp; RF Circuit Design:</b> Designing graph neural networks and transformers to automate analog and RF circuit design process including topology selection, layout optimization, and AI-guided circuit design (NeurIPS'25.)</li><li>• <b>Theory &amp; Applications of Generative Flow Networks:</b> Researching on the fundamentals and the applications of generative flow networks (GFlowNets) including sample efficiency, exploitation-exploration tradeoffs, and applications to hyperbolic geometry and federated learning (SIAM-SDM'25.)</li><li>• <b>Federated MRI Image Synthesis:</b> Designed the personalized &amp; federated image generation framework using generative adversarial networks for the first time in the literature (MIA'24, IEEE TMI'23, ISBI'23, ISMRM'23.)</li><li>• <b>Conformal Prediction for LLMs:</b> Developing uncertainty-aware LLM fine-tuning methods based on conformal prediction for close-ended and open-ended tasks to reduce the need of calibration data and extra post-hoc calibration step meanwhile improving learned representations.</li></ul> |                            |
| <b>TensorOpera AI (formerly FEDML AI)</b> , Research Scientist Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>May 2022 – Aug 2022</b> |
| <ul style="list-style-type: none"><li>• <b>FedGraphNN:</b> Developed a secure &amp; private data sharing platform for graph neural networks.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                            |
| <b>Turkcell Technologies</b> , Research & Development Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Oct 2018 - Dec 2020</b> |
| <ul style="list-style-type: none"><li>• <b>Spatiotemporal Forecasting with ConvLSTMs:</b> Developed and deployed a spatio-temporal forecasting algorithm with convolutional LSTM models.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |

## Honors & Awards

- **Top Reviewer** for NeurIPS'25.
- **Semi-finalist** for **Qualcomm Innovation Fellowship 2025**.  
*Abstract Title: AI Powered Revolution: Automating Analog and Radio Frequency Circuit Design*
- **2025 SIAM Travel Award** for SIAM-SDM'25 conference.
- **Travel & accommodation grant** for **PRAIRE AI 2019 summer school**.
- **Travel & accommodation grant** for **Deep|Bayes 2019 summer school (Acceptance rate:  $\leq 30\%$ )**.
- Funded through **5G and Beyond Joint Graduate Support Program** during M.Sc. studies by Information Technologies and Communication Authority (ICTA) (**First recipient**).
- **Full Scholarship** from Bilkent University for M.Sc. studies.
- **Research Excellence Award** by Bilkent University Electrical and Electronics Engineering Department.
- Bilkent University **High Honor student** during all semesters.
- Ranked in **top 2000** among 2M high school graduates in University Entrance Examinations (YGS/LYS), 2013.

### Published and Accepted Journal Publications

- [1] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “One model to unite them all: Personalized federated learning of multi-contrast MRI synthesis.” In *Medical Image Analysis*, Volume 94, May 2024, 103121.
- [2] G. Elmas, S. UH. Dar, Y. Korkmaz, **E. Ceyani**, B. Susam, M. Ozbey, **A.S Avestimehr**, T. Cukur. “Federated Learning of Generative Image Priors for MRI Reconstruction.” in *IEEE Transactions on Medical Imaging*, July 2023.

### Publications in Conference Proceedings & Workshops (Peer Reviewed)

- [1] A. Mehradfar, X. Zhao, X., Y. Huang, , **E. Ceyani**, Y. Yang, S. Han, H. Aghasi, & S. Avestimehr (2025). FALCON: An ML Framework for Fully Automated Layout-Constrained Analog Circuit Design. Accepted to NeurIPS’25,” 2025.
- [2] **E. Ceyani**, H. Xie, B. Buyukates, C. Yang, and **S. Avestimehr**. “FedGrAINS: Personalized Subgraph Federated Learning with Adaptive Neighbor Sampling.” In *SIAM International Conference on Data Mining (SDM25)*, 2025. (oral presentation, acceptance rate: 26.7%)
- [3] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “A Personalized Federated Learning Approach for Multi-Contrast MRI Translation,” in 31st annual meeting of International Society for Magnetic Resonance Imaging (ISMRM), Toronto, Canada, June 2023.
- [4] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “Personalized, Federated, And Unified MRI Contrast Synthesis,” in IEEE 20th International Symposium on Biomedical Imaging (ISBI), Virtual Conference, Apr. 2023. (Presented online)
- [5] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “pFLSynth: Personalized Federated Learning of Image Synthesis in Multi-Contrast MRI,” in NeurIPS Medical Imaging Meets, Virtual Conference (oral), Dec. 2022. (Presented online)
- [6] C. He\*, **E. Ceyani\***, K. Balasubramanian\*, M. Annavaram, and A.S. Avestimehr, “SpreadGNN: Serverless Multi-task Federated Learning for Graph Neural Networks,” accepted to *AAAI*, 2022. (acceptance rate of 15%) (co-first)
- [7] C. He\*, K. Balasubramanian\*, **E. Ceyani\***, C. Yang, H. Xie, L. Sun, L. He, L. Yang, P. S. Yu, Y. Rong, P. Zhao, J. Huang, M. Annavaram, and A.S. Avestimehr, “FedGraphNN: A Federated Learning System and Benchmark for Graph Neural Networks,” in *DPML workshop at ICLR and GNNSys workshop at MLSys*, 2021.
- [8] Ergen, Tolga, and **Emir Ceyani**. ”A highly efficient recurrent neural network architecture for data regression.” 2018 26th Signal Processing and Communications Applications Conference (SIU). IEEE, 2018.

## Professional Service

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- **Reviewer:** ICLR’24, NeurIPS’22,’23,’25(Main & Dataset and Benchmark Tracks), AAAI’26, Federated Learning with Graph Data (FedGraph2022-2023), CrossFL-MLSYS’22, IEEE Transactions of Neural Networks and Learning Systems, IEEE Transactions of Big Data
- **Organizer & Technical Committee Member:** 1st & 2nd International Workshops on Federated Learning with Graph Data, Technical Program Committee Member at the CrossFL Workshop, MLSYS2022
- **Summer Schools Attended:** LOGML’22 Summer School, ProbAI 2021 Summer School, PAISS Summer School’19, Deep—Bayes’19 Summer School

## Teaching Experience

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**University of Southern California**, Graduate Teaching Assistant

**Aug 2023 – May 2025**

*California, United States*

- Head Teaching Assistant for EE364: Introduction to Probability & Statistics for Electrical Engineers  
*Fall 2024 (Instructor: Salman Avestimehr), Spring 2025 (Instructor: Michael Neely)*
- Head Teaching Assistant for EE547: Applied Cloud Computing for Electrical Engineers  
*Fall 2023 (Instructor: Brandon Franzke)*

**ProbAI Summer School**, Voluntary Teaching Assistant

**Summer 2021**

*Remote*

**Bilkent University**, Graduate Teaching Assistant

**Sep 2018 – Dec 2020**

*Ankara, Türkiye*

- Grader for EEE443/543 - Neural Networks
- Teaching Assistant for EEE 424- Digital Signal Processing
- Teaching Assistant for EEE 321 - Signals & Systems
- Teaching Assistant for EEE 493/494 - Industrial Design Project I/II

## Technical Skills

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**Programming Languages:** Python, Matlab, Java, C++, Julia

**Developer Tools:**  $\LaTeX$ , VS Code, Jupyter, Inkscape

**Technologies/Frameworks:** Linux, PyTorch, PyTorch Geometric, TensorFlow, NumPy, Matplotlib, Pandas, Github

## Hobbies

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Playing Bass, 8-ball pool, Trading card games, Martial arts, Travelling, Culinary arts